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Adaptive Max Pressure Control of Network of Signalized Intersections

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Abstract:
The control of a road network of signalised intersections is considered. Variants of a decentralised universal feedback control policy (UFC) are evaluated and compared with an open loop traffic regulator periodically distributing the same actuation duration to each stage (set of permitted movements). At the beginning of each cycle and for a particular intersection, the Max-Pressure (MP) controller selects a stage to actuate as a function of the adjacent queues to that intersection. Vehicle movements are modelled as a store and forward (SF) queuing network. A controller stabilises a demand if the mean queue length remains bounded. The current MP version requires no knowledge of the demand while estimates turn probabilities. A section near LA comprised of 16 intersections and 76 links is employed. The network performance is appraised when regular traffic conditions, congestions or unforeseen feasible demand fluctuations take place. The conducted study assesses the potential benefits of MP and confirms its adaptability. Each control strategy is evaluated in terms of queue lengths, delays, travel times etc. by discrete event simulations confirming the provided theoretical guaranteed (queue) bounds of MP.

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Keywords: traffic-response signal control, stabilising policy, fixed-time and max-pressure controllers, feedback policies, store and forward queueing model, discrete-event simulation, DES, asynchronous systems, network calculus

1. INTRODUCTION

This study explores the potential of Max-Pressure feedback policy as an efficacious arterial traffic regulator. The system state is the vector of all queue lengths at all intersections the evolution of which is modelled as a controlled store-and-forward (SF) queuing network. The control which is a function of the current network state simultaneously activates one set of compatible *phases* or streams at pre-specified iid saturation flow rates. Each phase actuation permits vehicles to move from the input link of the phase to its output link. When a vehicle is discharged from a queue it travels to a randomly assigned destination at the downstream intersection according to a probability distribution. The exogenous demand at the input links is modelled as Poisson streams. A set of demands is said to be feasible, if there is a control that stabilises the queues, that is the time-average of every mean queue length is bounded. A major advantage of the MP version proposed in this paper and developed in P.Varaiya (2013), over other SF network control formulations Heydecker (2004), Aboudolas et al. (2009), Mirchandani and Head (2001) etc., consists in its decentralised character. At each intersection the control decision depends only on the queues adjacent to the node. Moreover, MP provably maximises throughput. If external arrivals and turn ratios are stationary, and if there is any policy that keeps all queues bounded, then max-pressure also keeps all queues bounded. Furthermore, no knowledge of the external arrivals is required although knowledge of turn ratios is expected, which can be estimated from the queue measurements. Interactions between vehicles and infrastructure form an asynchronous dynamical spatio-temporal system for which an event-driven approach is selected. Each employed strategy is quantified in terms of queue lengths, delays, trajectory travel times. This study is an extension of previous works of the authors in many aspects. First, the present version of the feedback MP strategy does not requires any knowledge of the turning probabilities. Second, in previous works an important variation of an external demand at a specific entry link was considered. The intersection instability under FT signals was then illustrated. Instead, the demand variation is now refined. Due to an in-depth analysis of the new network geometry, it is discussed how a small demand variation associated with pre-timed policies propagates instability to the entire network. Furthermore, the current research argues that a Fixed-Time policy stabilising the new demand in the related intersection is not enough to maintain stability in the entire network. Third, supposing that a demand variation can be detected, it is argued how the feedback time plans prevail system instability even when applied during the demand variation period. The fourth difference, concerns a new traffic context introduced in this work involving complete traffic blocking associated with some intersections. The impact of the control policy (FT, MP) before, while and after the traffic disturbance is studied. Fifth, delay metrics are presented in this work under different versions of feedback and open loop policies. Furthermore, an extended micro-analysis justifies the higher delays resulting by MP controller at a specific intersection while for the rest of the network the max-pressure performance are increased. The rest of the paper is organised as follows. §2 presents the problem formulation while §3

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provides different responses to this problem according to feedback and open loop approaches. Each control is evaluated by means of discrete event simulation. The choice of a DES modelling and the involvement of simulations are argued in §4. The network performance under regular traffic conditions and fixed demand is explored in §5. This section accentuates the need of a competent version of the employed feedback controller with an increased practical use. Traffic perturbations are involved prohibiting or reducing vehicle departs at a particular intersection during a given period. The decongestion process according to MP and Pre-timed policies is analysed in §6. Moreover, in §7 demand fluctuation is considered under regular traffic conditions and the network behaviour is quantified. §8 summarises the main conclusions of this study.

2. PROBLEM DESCRIPTION

A *stage* U_n is a binary matrix defining the simultaneously compatible movements (*phases*) of intersection n. Each queue corresponds to a distinct movement, so it is convenient to index a queue by a pair (m, n) in which m (n) is the incoming (outgoing) link index. For example, $q(l, m)$ is the number of vehicles in link l that are queued up at the corresponding intersection waiting to go to link m. Let $\mathcal{U}$ be the set of all possible *intersection stages*, $q(t) = \{q_{(l,m)}(t)\}$ the array of the queue lengths and $X(t)$ the system state at time t, then $X(t) = q(t)$. The optimisation horizon is divided into intervals or cycles of fixed width, each one comprising of T periods. Within each cycle, there exist $T - L$ available planning periods where $L < T$ represents the idle time corresponding to pedestrian movements, amber lights, etc. At the decision time t, we seek *stage* $u(t) = U, U \in \mathcal{U}$ and $\lambda_{u(t)}$ cycle proportion such that:

- $u(t)$ *stabilises* $X(t)$, that is
  - if $\tilde{c}(l, m)$, $r(l, m)$ denote the saturation flow rate and split ratio of phase (l, m) respectively and f_l is vehicle flow on link l, then
$$\tilde{c}(l, m) \geq f_l r(l, m) \ \forall (l, m) \text{ actuated by } U \quad (1)$$
- $\sum_{u \in \mathcal{U}} \lambda_u T + L = T$

3. TRAFFIC REGULATION

Hereafter, a standard *Fixed-time* (FT) approach and two versions of the Max-Pressure controller are presented.

3.1 Open Loop Pre-Timed Control

A *Fixed-time* control is a periodic sequence, $\{\lambda_{U^i},\ U^i \in \mathcal{U}\}$, actuating each *stage* $u(t) = U^i$, $U^i \in \mathcal{U}$ for a fixed duration of $\lambda_{U^i} T$ time periods during every cycle T. If L denotes the total red clearance period per cycle, then $\sum_i \lambda_i = 1 - L/T$.

3.2 Max-Pressure Feedback Control

The *pressure* $w(q(t), U)$ exerted at time t by a stage U to the intersection of state $q(t)$ is defined by

$$w(q(t), U) = \sum_{(l,m)} \varsigma(l, m)(t) S \circ U(l, m)(t), \quad (2)$$

where ς is defined by

$$\varsigma(l, m)(t) = q_{(l,m)}(t) - \sum_{p \in \mathcal{O}(m)} r(m, p) q_{(m,p)}(t). \quad (3)$$

At any decision moment t, the selected MP control stage U^* is the one maximising the exerted pressure at state $q(t)$,

$$U^*(q)(t) = \text{argmax}\{w(q(t), U), U \in \mathcal{U}\}. \quad (4)$$

One may observe that Max-Pressure may be implemented incrementally: if a new intersection is added to the network, the max pressure policy for the original network does not change. In the centralised control of the cited studies in §1 an expansion of the controlled network leads to changes in the policy of all intersections. As presented in P.Varaiya (2013) the MP controller is robust in the sense that if any controller can keep queues bounded, so can the max-pressure. From the proof one gains the intuition that the max-pressure controller attempts at each t to minimise $|q(t + 1)|^2$ given $q(t)$.

3.3 Adaptive Max-Pressure (AMP)

In the MP stage decision, (relation 4), the turning probabilities of the related to the intersection movements are required. However, turn routings can also be on-line estimated (by queue measurements). If $a(l, i)(t)$ denotes vehicle arrivals at phase (l, i) of link l during period t where i is a possible destinations from link l, then estimated turn ratio value $\hat{r}(l, m)$ of phase (l, m), can be computed by

$$\hat{r}(l, m) = \frac{\sum_t a(l, m)(t)}{\sum_k \sum_t a(l, k)(t)} \quad (5)$$

Using equation (5) in (4) the AMP actuation stage results. If $\hat{r}(l, m)$ is consistent, AMP is stabilising.

4. PERFORMANCE EVALUATION FRAMEWORK

Traffic management forms an asynchronous structure the dynamics of which cannot be precisely described by closed-form expressions. Formally, a Discrete Event System (DES) is a dynamical system with discrete states whose transitions are triggered by events, that is the behaviour of a DES is ruled only by occurrences of different types of events over time rather than by fixed time steps. Although time evolves in between occurrences, that does not directly affect the system. Furthermore, at least some of the evolved variables are discrete. The event-driven nature of a DES forces for new mathematical approaches for modelling and analysis, since differential/difference equations (developed for the analysis of time-driven systems) no longer form an adequate setting. Additionally, transitional choices have to be made in DES, either internally by the system itself using some chance mechanism or from the

external world. Simulation means consist a faithful way to describe the DES dynamics and allow for an efficacious performance evaluation. For the study of the arterial management a micro-meso discrete event decision tool, PointQ, is employed involving events on vehicle arrivals, departures and signal actuation. At any time, the sum of the vehicle storage capacity of all queues of an internal link is bounded by the link capacity. Each "point" queue discharges at the saturation flow rate when the signal is green, and is fed by exogenous arrivals and/or by vehicles that are routed from other queues. Performance measures are calculated such as travel times, queue lengths, delays etc.

5. EXPERIMENTING TRAFFIC STATE

A simulation study of a road network near Los Angeles with 16 signalised intersections, 73 links (22 entry links, 24 exit ones), and 179 turn movements, hence 179 queues is now presented. Figures 1 & 2 show a map of the study site and its representation as a directed graph, respectively. Exogenous demands at the input links are modelled as stationary Poisson streams and intersections regulated by the above described controls (§3). Approximatively 43, 033 vehicles are generated over three hours. Link travel times are stochastic depending on the free flow speed and the current link state. Time granularity is taken equal to 0.1 seconds and each implementation duration is of 10,800 seconds (3 hours). Turning ratios are estimated every 15 minutes.

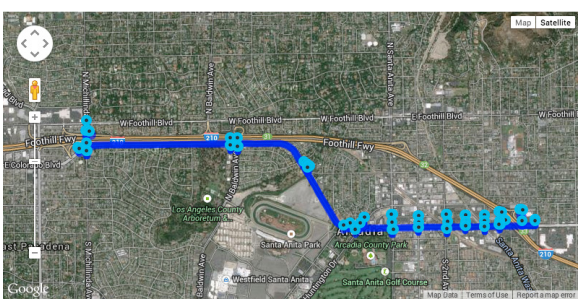

Fig. 1. Huntington-Colorado arterial (near I-210), CA.

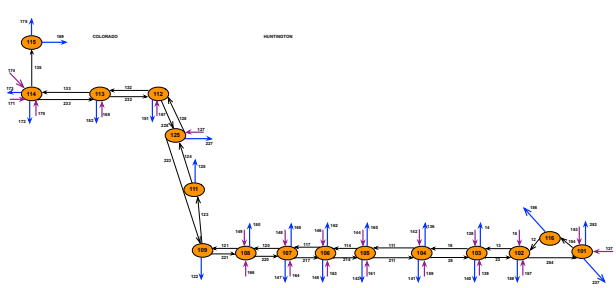

Fig. 2. Huntington-Colorado graph

Next, we quote results of the network performance under FT and MP control policies. Within the purpose of an accurate comparison, all implementations employe an identical demand (intensity-geometry).

5.1 Pre-Timed Control with offsets

A stabilising fixed time (with offsets) control is considered (for most nodes, the cycle duration is of 120 secs and the idle time is of 2 secs per cycle). The evolution of the sum of all the network is depicted in Figure 3. As expected the mean queue is bounded for the related demand and the associated fixed time actuation durations.

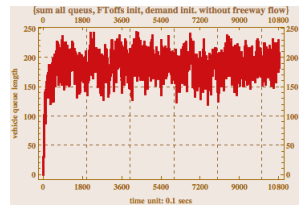

Fig. 3. Sum queues-FT with offsets control

5.2 MP Adaptive Control

As discussed in §3 since there exists a stable FT control, the MP controller should also be stable. Figure 4 illustrates this property under MP policy and while the control decision is revised eight times per cycle. In comparison with FT, the MP maintains lower values for the sum of all the network queues.

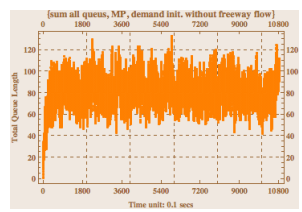

Fig. 4. Sum queues-MP (adaptive) 8 revisions per cycle

5.3 Frequency of the control revision and network behaviour

A MP control is parametrised by the number of decision times within each cycle that the duration can be adjusted. The theory of MP developed in P.Varaiya (2013), shows that the queue length resulting from MP control is inversely proportional to the number of decisions per cycle. This property is presented in Figure 5. Increased frequency of control updates leads to diminished network queues. Although this decrease is significant from 2 to 8 control revisions per cycle it becomes less important from 8 to 10 decision updates. Could one claim that often control revision is an efficient issue for diminishing the total network queue ?

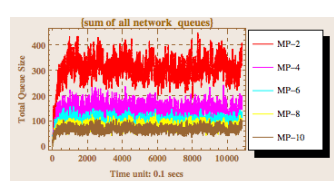

Fig. 5. Sum queues, MP multiple decisions

Stage changes incur loss due to the idle time interfering between two distinct stages. It is meaningless to reduce queues by increasing the MP control updates. The MP-Practical policy described below achieves equivalent effects while weakens inconveniences resulting from frequent stage switches.

5.4 MP - Practical Control, MP-Pract

A MP-Pract controller permits similar network performance to a MP one with frequent decision revisions but with a reduced cost. Thus, one can evaluate *MP stage* U^* frequently and implement the new selected *MP stage* change only if significantly larger pressure w,

$$\max_U w(U, q(t)) \geq (1+\eta) w(U^*, q(t)). \tag{6}$$

Figure 6 illustrates the sum of the network queues when 2, 4, 6, 8, 10 revisions per cycle and parameter $\eta = 0.2$. Comparison of Figures 5, 6 confirms stability and close network performance. Figures 7, 8 present the network performance under MP-Pract with $2, 4, 6, 8, 10$ control updates per cycle and $\eta = 0.4, 1.2$.

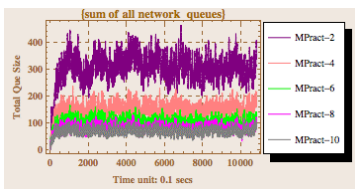

Fig. 6. Sum of queues, MP-Practical, parameter $\eta = 0.2$

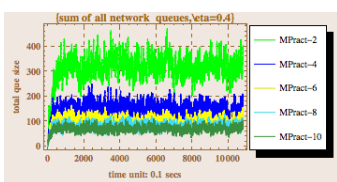

Fig. 7. Sum of queues, MP-Practical, parameter $\eta = 0.4$

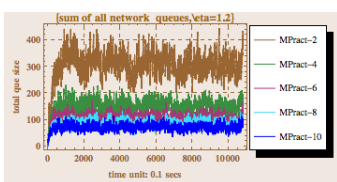

Fig. 8. Sum of queues, MP-Practical, parameter $\eta = 1.2$

Table 1 resumes the number of stage switches when MP-Pract with 2 to 10 evaluations per cycle and when parameter $\eta = 0.2, 1.2$, for intersections 101 and 102. Thus, for node 101, when MP with 4 decision per cycle (MP4), the control is evaluated 956 times while when MP-Pract with 10 revisions per cycle and $\eta = 1.2$, 901 control switches are resulted. Moreover the network performance when 10 control updates are much higher than those of MP4.

5.5 Performance Comparison

For each employed policy (FT and MP versions) the probability of each value taken by the sum of all the network queues is computed and depicted in Figure 9. Thus, under MP4 a queue size of 150 vehicles is of the most probable value while when MP10 this value is reduced to 50 vehicles approximatively.

Node ID 10001 **Nb Eval.**	**Nb Eval. MP**	**Nb Eval. MPract** *η : 0.4*	**Nb Eval. MPract** *η : 1.2*	**Nb Switch. MP**	**Nb Switch. MPract** *η : 1.2*
2	480	480	480	240	239
4	956	944	916	475	434
6	1422	1395	1346	699	619
8	1877	1810	1738	914	772
10	2314	2211	2108	1111	901
10002					
2	360	360	360	180	180
4	708	711	701	347	340
6	1033	1021	1022	491	481
8	1309	1303	1311	586	588
10	1537	1537	1544	632	639

Table 1. MP-Pract, $\eta = 0.2, 1.2$,

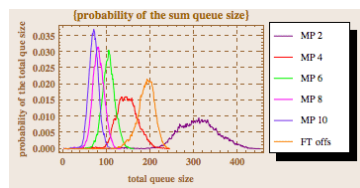

Fig. 9. Probability of the size of the sum of all the network queues

6. TRAFFIC PERTURBATION

6.1 Car Blocking

The case when the accessibility of an intersection node, is reduced (potential incident etc.) or prohibited (complete intersection blocking) is now considered. This section argues how the network evolves within the new traffic context for both FT and or MP control policies.

The study horizon of $10,800$ secs is decomposed into three time periods: period 1: $[0, 1000)$, period 2: $[1000, 4600)$ and period 3: $[4600, 10800)$.

During periods 1 and 3 all intersection phases are accessible while during period 2 at most two vehicles per cycle depart from intersection 103 when MP control while 6 vehs/cycle depart under FT (instead of 118 vehs/cycle).

6.2 Scenario 1: FT and MP4 controllers

The network is ruled by a FT control during all time periods. Alternatively a MP4 policy is applied during period 3. As illustrated in Fig.10 most vehicles fail to depart during period 2 (blue curve controlled by FT policy). As expected the network becomes unstable. Period 1 is represented by the red part while the green curve corresponds to period 3 when the FT policy is applied. The cyan curve represents the network behaviour when a MP4 strategy rules as soon as node 103 becomes accessible. One observes that *soon* MP diminishes the sum of the network queues more than FT.

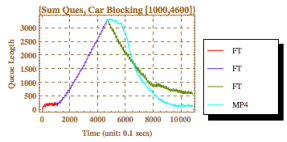

Fig. 10. Sum queues-FT-MP4, congestion

6.3 Scenario 2: MP4 and MP8 controllers

Figure 11 presents the network evolution when a MP4 policy is employed (red, green and blue curves) from $t = 0$ to $t = 10,800$. secs The orange curve corresponds to the case when a MP8 control rules just after the end of the blocking period.

Figures 10, 11 show improved network performance under MP policy during the entire period $[0, 10800)$ although no vehicle departs node 103 during period 2 under MP.

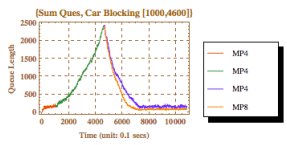

Fig. 11. Sum queues-MP4-MP8, congestion

6.4 Micro Analysis - Nodes 103, 102

Distinct queues at node 103 are examined as well upstream queues at intersection 102. Phases $(103, 108)$, $(138, 140)$ correspond to node 103 where $(105, 103)$ is a phase of node 102 discharging vehicles towards node 103. Figures 12, 13, 14 present the evolution of queues $q(103, 108)$, $q(138, 140)$ and $q(105, 103)$ during the congestion period 2 and FT or MP4. Irregularity at node 103 affects not only local movements but also propagates congestion to the upstream queues.

Note that links 138 and 105 are entry links and a infinite vehicle storage capacity is considered. A finite capacity is associated with the internal network link 103 equal to 50 vehicles. When this value is reached the related link is saturated. Again, one observes higher network performance under the feedback MP policy.

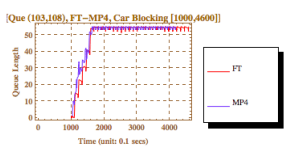

Fig. 12. $q(103, 108)$: FT-MP4, period 2

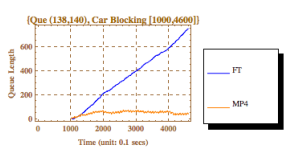

Fig. 13. $q(138, 140)$: FT-MP4, period 2

Delays related to queues $q(103, 108)$ and $q(138, 140)$ are now examined.

Figure 15 presents the delay evolution at phase $(103, 108)$. The sequence of {brown, cyan, yellow} curves represents

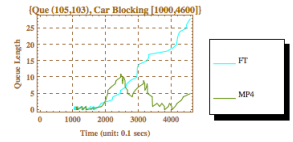

Fig. 14. $q(105, 103)$: FT-MP4, period 2

the case when FT control is employed during all periods. The sequence of {brown, cyan, green} curves corresponds to the case when pre-timed control is employed during periods 1 and 2 while a MP4 policy is employed by the end of the congestion period. One observes that during most part of period 3 FT delays are smaller. This initially surprising result is justified, by the MP algorithm described in §3. Loosely speaking, MP actuates stages whose movements have the higher queues while it takes into consideration the state of the downstream queues at the decision time.

Two competitive *stages* correspond at intersection 103, one actuating $q(103, 108)$, $q(208, 203)$, the other actuating $q(138, 140)$, $q(139, 104)$. Internal link 208 is of finite capacity. As Fig. 13 illustrates $q(138, 140)$ becomes unbounded during period 2 and consequently takes much higher values than queues of stage 1. MP applied at the end of period 2, will prioritise the stage actuating $q(138, 140)$ since this movement is in a critical state. Figure 16 presents the delay evolution at queue $q(138, 140)$. MP4 policy decongests much faster $q(138, 140)$ rather than FT does.

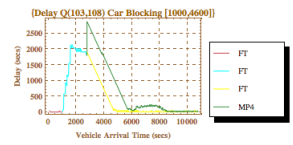

Fig. 15. Delay $q(103, 108)$: FT-MP4, congestion

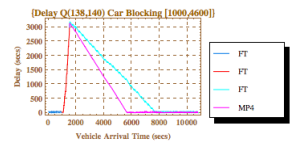

Fig. 16. Delay $q(138, 140)$: FT-MP4, congestion

Figure 17 illustrates the delay evolution of $q(103, 108)$ under MP4 and MP8 control. Comparison of Figures 15, 17 shows that MP control not only keeps lower delays but during the decongestion period 3, MP recovers the system to the regular delay values, faster than FT.

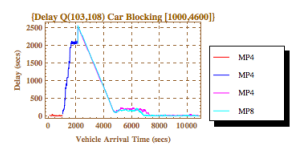

Fig. 17. Delay $q(103, 108)$: MP4-MP8, congestion

7. INFLUENCE OF THE DEMAND INTENSITY

The case of the network performance under FT and MP policies associated with a demand variation is now studied. In the employed scenario, demand (slightly) fluctuates

at intersection 101. Thus, during $[0, 12,000)$ the (vector of) demand intensity is of d_1 and becomes d_2 during $[12,000, 21,000)$. Let $\{\lambda^1, d_1\}$ be a stabilising Pre-Timed control accommodating demand d_1.

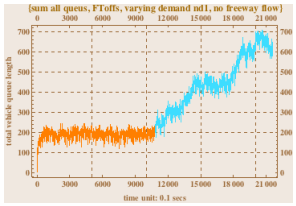

Fig. 18. Network evolution under $\{\{\lambda^1, d_i\},\ i = 1, 2\}$

Fig. 18 illustrates the case when the same FT control is applied for both periods $[0, 12,000)$ and $[12,000, 21,000)$. Before the demand fluctuation the network remains stable (mean queue length bounded, orange curve). When demand becomes d_2 these actuation times fail to stabilize the network (cyan curve).

Suppose that a FT duration is recomputed for node 102 stabilising its new demand. The new control $\{\lambda_2, d_2\}$ (employing the same actuations as vector λ^1 for any other node) is applied during period $[12,000, 21,000)$. Figure 19 depicts the new network performance. Although the FT accommodates the new demand at node 101 the network remains unstable (blue curve). Instability is propagated to the following nodes, internal link flows are affected by the demand variation. Figure 20 illustrates the evolution of the adjacent que $q(108, 136)$ to node 103 under $\{\lambda_2, d_2\}$ (d_2 maintains the same external demand at node 103).

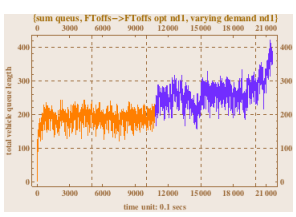

Fig. 19. Network evolution when $\{\{\lambda^1, d_1\}, \{\lambda_2, d_2\}\}$

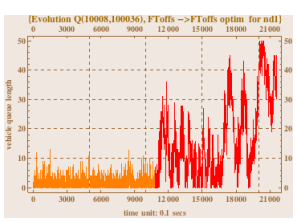

Fig. 20. $q(108, 136)$ under $\{\{\lambda^1, d_1\}, \{\lambda_2, d_2\}\}$

Finally a stabilising FT control $\{\lambda^2, d_2\}$ is recomputed for the entire network and for the new demand d_2. Figure 21 shows the network stability.

As illustrated in Figure 22 a MP policy applied during $[12,000, 21,000)$ automatically adapts to the new (feasible) demand and stabilises the network. Moreover, it maintains lower values for the sum of the network queues regarding the corresponding ones under FT.

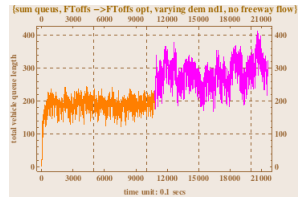

Fig. 21. Sum queues under $\{\{\lambda^1, d_1\}, \{\lambda^2, d_2\}\}$

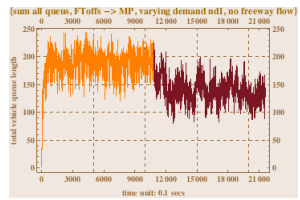

Fig. 22. Sum queues under $\{\{\lambda^1, d_1\}, \{MP6, d_2\}$

8. DISCUSSION

The potential of the Max-Pressure (MP) feedback policy as an efficacious arterial traffic regulator is developed. Regular and congestion traffic conditions are considered and the network performance is quantified (some of the considered performance metrics are queue lengths and delays) under MP versions and Pre-Timed control. Feasible demand fluctuations confirm the adaptive character of MP preserving the network stability. The usefulness of MP-Pract is examined. Provided that the majority of the traffic controller in the US are FT, the effectiveness of MP-Pract worths to be explored. The traffic management constitues a dynamical asynchronous system for which a DES model is employed associated with discrete event simulations. PointQ is an event-approach decision tool utilised during this study for the evaluation of the network performance according to different strategies. A MP version estimating queue lengths is under developement.

REFERENCES

Aboudolas, K., Papageorgiou, M., and E.Kosmatopoulos (2009). Store-and-forward based methods for the signal control problem in large-scale congested urban road networks. *Transportation Research Part C-Emerging Technologies*, 17, 163–174.

Allsop, R.E. (1971). Delay-minimizing settings for fixed-time traffic signals at a single road junction. *IMA Journal of Applied Mathematics, No. 2*, 8, 164–185.

Baccelli.F, Cohen.G, Olsder.J, and Quadrat.G (1992). Synchronization and linearity – an algebra for discrete event systems. *Wiley, New-York*.

Gomes.G, Horowitz.R., Kurzhanskiy.A., Kwon.J, and Varaiya.P (2008). Behaviour of the cell transmission model and effectiveness of ramp metering.

Heydecker, B. (2004). Objectives, stimulus and feedback in signal control of road traffics. *Journal of Intelligent Transportation Systems*, 8, 63–76.

Mirchandani, P. and Head, L. (2001). A real-time traffic signal control system: Architecture, algorithms, and analysis. *Transportation Research Part C-Emerging Technologies*, 9(6), 415–432.

P.Varaiya (2013). Max pressure control of a network of signalized intersections. *Springer*.